Tech Dessert #11: RF Exposure Safety

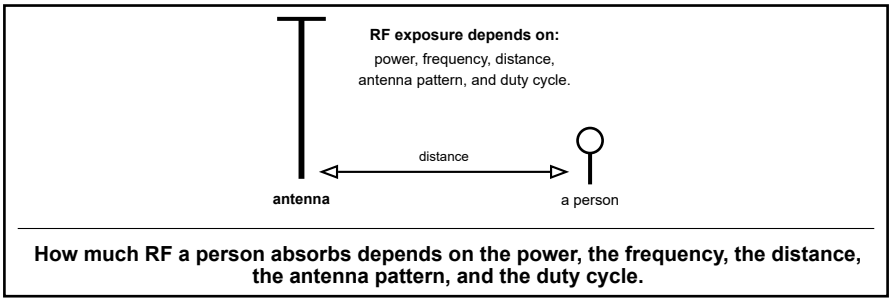
Companion reference for meeting 11 -- what RF radiation is, what affects exposure, and how to keep your station safe and legal.

Group T0C · 13 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert is about transmitting safely around people. The good news first: radio energy is **non-ionizing** – but it still must be handled with care. Read the table, study the diagram, then drill the question bank.

RF Exposure Safety

Topic	What to know	Q
What kind of radiation radio signals are	**Non-ionizing** radiation.	T0C01
How RF differs from ionizing radiation	RF does **not have enough energy** to cause chemical changes or damage DNA.	T0C12
What factors affect RF exposure	The field's power and frequency, the distance to the person, and the antenna's radiation pattern – **all of these**.	T0C04
Why exposure limits vary with frequency	The body **absorbs more RF energy at some frequencies** than others.	T0C05
Band with the lowest permissible exposure	**50 MHz.**	T0C02
What "duty cycle" means	The **percentage of time** a transmitter is actually transmitting.	T0C11
Why duty cycle matters for RF safety	It affects the **average** exposure.	T0C10
Effect of duty cycle dropping from 100% to 50%	The allowable power density **doubles** (increases by a factor of 2).	T0C03
Hazard of touching an antenna while transmitting	An **RF burn** to the skin.	T0C07
How to reduce RF exposure	**Relocate the antenna** farther from people.	T0C08
How to show your station complies	OET Bulletin 65 calculation, computer modeling, or measurement – **all of these**.	T0C06
Staying in compliance over time	**Re-evaluate** the station whenever the transmitter or antenna system is changed.	T0C09
Who is responsible for keeping people safe	The **station licensee**.	T0C13



How much RF a person absorbs depends on the power, the frequency, the distance, the antenna pattern, and the duty cycle.

Remember: radio energy is **non-ionizing** – it cannot damage DNA the way X-rays can – but a strong field still heats tissue, and touching a transmitting antenna gives an **RF burn**. Exposure depends on **power, frequency, distance, antenna pattern, and duty cycle**. The simplest fix for too much exposure is **more distance** – relocate the antenna. And the **station licensee** is the one responsible for keeping everyone within the limits.

Dessert 11 Question Bank (T0C)

T0C01. What type of radiation are radio signals?
A. Gamma radiation
B. Ionizing radiation
C. Alpha radiation
D. Non-ionizing radiation

T0C02. Which of the following bands has the lowest maximum permissible exposure for RF safety?
A. 3.5 MHz
B. 50 MHz
C. 440 MHz
D. 1296 MHz

T0C03. How does the allowable power density for RF safety change if duty cycle changes from 100 percent to 50 percent?
A. It increases by a factor of 3
B. It decreases by 50 percent
C. It increases by a factor of 2
D. There is no adjustment allowed for lower duty cycle

T0C04. What factors affect the RF exposure of people near an amateur station antenna?
A. Frequency and power level of the RF field
B. Distance from the antenna to a person
C. Radiation pattern of the antenna
D. All these choices are correct

T0C05. Why do exposure limits vary with frequency?
A. Lower frequency RF fields have more energy than higher frequency fields
B. Lower frequency RF fields do not penetrate the human body
C. Higher frequency RF fields are transient in nature
D. The human body absorbs more RF energy at some frequencies than at others

T0C06. Which of the following is an acceptable method to determine whether your station complies with FCC RF exposure regulations?
A. By calculation based on FCC OET Bulletin 65
B. By calculation based on computer modeling
C. By measurement of field strength using calibrated equipment
D. All these choices are correct

T0C07. What hazard is created by touching an antenna during a transmission?
A. Electrocution
B. RF burn to skin
C. Exposure to ionizing radiation
D. All these choices are correct

T0C08. Which of the following actions can reduce exposure to RF radiation?
A. Relocate antennas
B. Relocate the transmitter
C. Increase the duty cycle
D. All these choices are correct

T0C09. How can you make sure your station stays in compliance with RF safety regulations?
A. By informing the FCC of any changes made in your station
B. By re-evaluating the station whenever an item in the transmitter or antenna system is changed
C. By making sure your antennas have low SWR
D. By using only Underwriter Laboratories approved transmitting equipment

T0C10. Why is duty cycle one of the factors used to determine safe RF radiation exposure levels?
A. It affects the average exposure to radiation
B. It affects the peak exposure to radiation
C. It takes into account the antenna feed line loss
D. It takes into account the thermal effects of the final amplifier

T0C11. What is the definition of duty cycle during the averaging time for RF exposure?
A. The difference between the lowest and highest power output of a transmitter
B. The difference between the PEP and the average power output of a transmitter
C. The percentage of time that a transmitter is transmitting
D. The percentage of time that a transmitter is not transmitting

T0C12. How does RF radiation differ from ionizing radiation (radioactivity)?
A. RF radiation does not have sufficient energy to cause chemical changes in cells and damage DNA
B. RF radiation can only be detected with an RF dosimeter
C. RF radiation is limited in range to a few feet
D. RF radiation is perfectly safe

T0C13. Who is responsible for ensuring that no person is exposed to RF energy above the FCC exposure limits?
A. The FCC
B. The station licensee
C. Anyone who is near an antenna
D. The local zoning board

Answer Key

T0C01: D	T0C02: B	T0C03: C	T0C04: D	T0C05: D
T0C06: D	T0C07: B	T0C08: A	T0C09: B	T0C10: A
T0C11: C	T0C12: A	T0C13: B		